ENV 797 - Time Series Analysis for Energy and Environment Applications | Spring 2026

Forecasting Competition Final Report

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Setting R code chunk options

GitHub Repository: ShenWu_GithubRepo

Overview

This report implements and evaluates **five Prophet-based forecasting models** for daily electricity demand, forecasting July 2011 using data from January 2005 to June 2011.

Model	Regressors	Tuned	CV Window	Selection Metric
`Submission_Prophet_Tuned`	Yes	Yes	5 years	MAPE
`Sub_M4_Prophet_NoReg_Baseline`	No	Yes	5 years	RMSE
`Sub_M4_Prophet_NoReg_Baseline (v2)`	No	Yes	5 years	MAPE
`Submission_Prophet_Tuned_NEW`	No	Yes	4 years	MAPE
`Submission_Prophet_Tuned_Newest51`	No	Yes	5 years	MAPE

All models use daily (site-level averaged) data with walk-forward cross-validation (1-month horizon, 1-month skip). All five models are tuned via grid search.

The hyperparameters searched across all models are: `prior_scale_changepoints`, `prior_scale_seasonality` , `prior_scale_holidays` (active grid; full sweep values listed in Section 10 and commented out to make compiling easier). Structural parameters `n.changepoints`, `changepoint.range`, and `seasonality.mode` are fixed via `set_engine()`.

Additionally, we tested different loss terms like MAE, RMSE and MAPE (We discovered the MAPE metric late!)

We relied on Prophet for all our best models as it is a very flexible Bayesian Model that generally has strong performance for seasonal data and worked very well. We found it easy to use as it has a good codebase and extensive documentation and online help.

We used LLMs to troubleshoot and clean up our code as well as to generate the hyper parameter sweep setup.

1.2 Load Packages

```
library(lubridate)
library(ggplot2)
library(forecast)
```

```r
library(Kendall)
library(tseries)
library(outliers)
library(tidyverse)
library(smooth)
library(zoo)
library(kableExtra)
library(readxl)
library(tidymodels)
library(modeltime)
library(timetk)
library(tune)
library(dials)
```

---

2. Data Import

The dataset comes from the Kaggle competition platform. There are three files: hourly electricity demand (load), hourly temperature, and hourly relative humidity from January 2005 to June 2011. The load file has one row per day with 24 columns (h1 to h24) representing hourly readings. The temperature and humidity files have one row per hour with readings from 28 weather stations.

```r
load_data <- read_excel("./Data/load.xlsx")
temp_data <- read_excel("./Data/temperature.xlsx")
rh_data   <- read_excel("./Data/relative_humidity.xlsx")

head(load_data)
```

```
## # A tibble: 6 x 26
##   meter_id date                  h1    h2    h3    h4    h5    h6    h7    h8
##   <chr>    <dttm>              <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0072     2005-01-01 00:00:00  3072  2911  2784  2768  2832  2904  3064  3192
## 2 0072     2005-01-02 00:00:00  2408  2328  2311  2328  2400  2552  2856  3272
## 3 0072     2005-01-03 00:00:00  2575  2528  2496  2608  2825  3376  3760  3672
## 4 0072     2005-01-04 00:00:00  2177  2016  1951  1913  2080  2624  3328  2896
## 5 0072     2005-01-05 00:00:00  1912  1768  1697  1720  1880  2472  3208  2888
## 6 0072     2005-01-06 00:00:00  2104  1976  1920  1952  2167  2736  3432  3096
## # i 16 more variables: h9 <dbl>, h10 <dbl>, h11 <dbl>, h12 <dbl>, h13 <dbl>,
## #   h14 <dbl>, h15 <dbl>, h16 <dbl>, h17 <dbl>, h18 <dbl>, h19 <dbl>,
## #   h20 <dbl>, h21 <dbl>, h22 <dbl>, h23 <dbl>, h24 <dbl>
```

```r
cat("Load data rows:", nrow(load_data), "\n")
```

```
## Load data rows: 2372
```

```r
cat("Temp data rows:", nrow(temp_data), "\n")
```

```
## Temp data rows: 56922
```

```r
cat("RH data rows:",  nrow(rh_data),   "\n")
```

```
## RH data rows: 56921
```

3. Data Wrangling

3.1 Daily Load (averaged across 24 hourly columns)

```r
load_daily <- load_data %>%
  mutate(date = ymd(date)) %>%
  rename(Date = date) %>%
  pivot_longer(cols = h1:h24, names_to = "hour", values_to = "load") %>%
  group_by(Date) %>%
  summarise(daily_mean_load = mean(load, na.rm = TRUE))

cat("NAs in daily load:", sum(is.na(load_daily$daily_mean_load)), "\n")
```

```
## NAs in daily load: 0
```

```r
summary(load_daily)
```

```
##       Date             daily_mean_load
##  Min.   :2005-01-01   Min.   :2247
##  1st Qu.:2006-08-16   1st Qu.:2798
##  Median :2008-03-31   Median :3506
##  Mean   :2008-03-31   Mean   :3629
##  3rd Qu.:2009-11-14   3rd Qu.:4211
##  Max.   :2011-06-30   Max.   :7897
```

3.2 Daily Temperature (averaged across 28 stations)

```r
# Single-column daily average (used as regressor in M1)
temp_daily <- temp_data %>%
  mutate(date = ymd(date)) %>%
  rename(Date = date) %>%
  pivot_longer(cols = t_ws1:t_ws28, names_to = "station", values_to = "temp") %>%
  group_by(Date) %>%
  summarise(daily_mean_temp = mean(temp, na.rm = TRUE)) %>%
  filter(!is.na(Date))

# All-station daily averages (used for future regressor construction)
temp_daily_allsites <- temp_data %>%
  mutate(date = ymd(date)) %>%
  rename(Date = date) %>%
  filter(!is.na(Date)) %>%
  group_by(Date) %>%
  summarise(across(t_ws1:t_ws28, ~ mean(.x, na.rm = TRUE)))

summary(temp_daily)
```

```
##       Date            daily_mean_temp
##  Min.   :2005-01-01   Min.   :21.39
##  1st Qu.:2006-08-16   1st Qu.:48.53
##  Median :2008-03-31   Median :61.90
##  Mean   :2008-03-31   Mean   :60.44
##  3rd Qu.:2009-11-14   3rd Qu.:74.12
##  Max.   :2011-06-30   Max.   :87.09
```

3.3 Daily Relative Humidity (averaged across 28 stations)

```r
# Single-column daily average (used as regressor in M1)
rh_daily <- rh_data %>%
  mutate(date = ymd(date)) %>%
  rename(Date = date) %>%
  pivot_longer(cols = rh_ws1:rh_ws28, names_to = "station", values_to = "rh") %>%
  group_by(Date) %>%
  summarise(daily_mean_rh = mean(rh, na.rm = TRUE)) %>%
  filter(!is.na(Date))

# All-station daily averages
rh_daily_allsites <- rh_data %>%
  mutate(date = ymd(date)) %>%
  rename(Date = date) %>%
  filter(!is.na(Date)) %>%
  group_by(Date) %>%
  summarise(across(rh_ws1:rh_ws28, ~ mean(.x, na.rm = TRUE)))

summary(rh_daily)
```

```
##       Date            daily_mean_rh
##  Min.   :2005-01-01   Min.   :27.10
##  1st Qu.:2006-08-16   1st Qu.:61.12
##  Median :2008-03-31   Median :69.83
##  Mean   :2008-03-31   Mean   :68.56
##  3rd Qu.:2009-11-14   3rd Qu.:77.01
##  Max.   :2011-06-30   Max.   :94.95
```

3.4 Combined Daily Dataset (with gap-filling)

```r
daily_data <- load_daily %>%
  left_join(temp_daily, by = "Date") %>%
  left_join(rh_daily,   by = "Date")

# Full consolidated frame (all station columns) with forward/backward fill
df_final <- daily_data %>%
  left_join(temp_daily_allsites, by = "Date") %>%
  left_join(rh_daily_allsites,   by = "Date") %>%
  arrange(Date) %>%
  fill(everything(), .direction = "downup")

head(df_final)
```

```
## # A tibble: 6 x 60
##   Date       daily_mean_load daily_mean_temp daily_mean_rh t_ws1 t_ws2 t_ws3
##   <date>               <dbl>           <dbl>         <dbl> <dbl> <dbl> <dbl>
## 1 2005-01-01           3108.            53.6          76.7  50.2  53.3  49.7
## 2 2005-01-02           3068.            53.8          80.5  55.3  51.6  49.0
## 3 2005-01-03           3061.            55.9          81.2  55.8  56.5  51.4
## 4 2005-01-04           2566.            61.7          74.8  58.0  63.3  57.1
## 5 2005-01-05           2496.            60.4          76.1  62.5  63.1  52.5
## 6 2005-01-06           2735.            62.0          78.0  60.5  66.1  54.5
## # i 53 more variables: t_ws4 <dbl>, t_ws5 <dbl>, t_ws6 <dbl>, t_ws7 <dbl>,
## #   t_ws8 <dbl>, t_ws9 <dbl>, t_ws10 <dbl>, t_ws11 <dbl>, t_ws12 <dbl>,
## #   t_ws13 <dbl>, t_ws14 <dbl>, t_ws15 <dbl>, t_ws16 <dbl>, t_ws17 <dbl>,
## #   t_ws18 <dbl>, t_ws19 <dbl>, t_ws20 <dbl>, t_ws21 <dbl>, t_ws22 <dbl>,
## #   t_ws23 <dbl>, t_ws24 <dbl>, t_ws25 <dbl>, t_ws26 <dbl>, t_ws27 <dbl>,
## #   t_ws28 <dbl>, rh_ws1 <dbl>, rh_ws2 <dbl>, rh_ws3 <dbl>, rh_ws4 <dbl>,
## #   rh_ws5 <dbl>, rh_ws6 <dbl>, rh_ws7 <dbl>, rh_ws8 <dbl>, rh_ws9 <dbl>, ...
```

```r
summary(df_final)
```

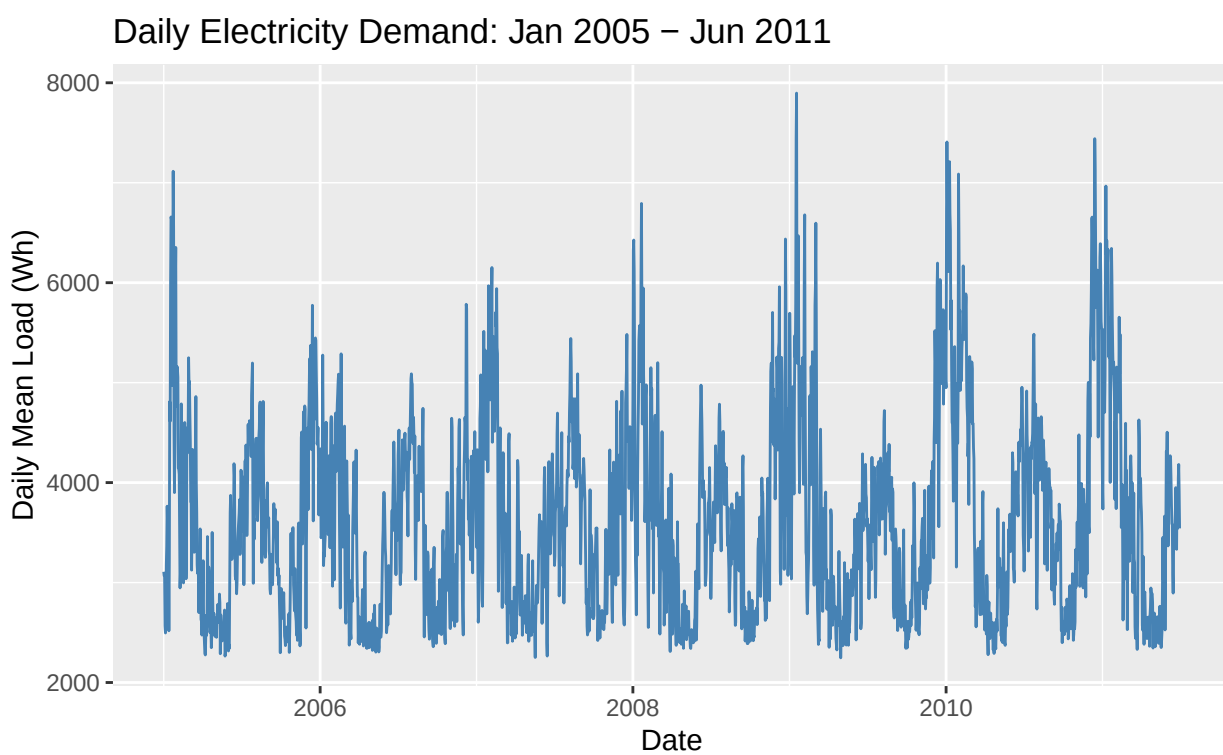
```
##       Date            daily_mean_load daily_mean_temp daily_mean_rh
##  Min.   :2005-01-01   Min.   :2247    Min.   :21.39   Min.   :27.10
##  1st Qu.:2006-08-16   1st Qu.:2798    1st Qu.:48.53   1st Qu.:61.12
##  Median :2008-03-31   Median :3506    Median :61.90   Median :69.83
##  Mean   :2008-03-31   Mean   :3629    Mean   :60.44   Mean   :68.56
##  3rd Qu.:2009-11-14   3rd Qu.:4211    3rd Qu.:74.12   3rd Qu.:77.01
##  Max.   :2011-06-30   Max.   :7897    Max.   :87.09   Max.   :94.95
##      t_ws1           t_ws2           t_ws3           t_ws4
##  Min.   :23.29   Min.   :21.00   Min.   :12.54   Min.   :21.71
##  1st Qu.:48.04   1st Qu.:48.16   1st Qu.:43.58   1st Qu.:48.92
##  Median :62.21   Median :62.69   Median :56.54   Median :62.21
##  Mean   :60.86   Mean   :60.87   Mean   :55.18   Mean   :60.83
##  3rd Qu.:75.08   3rd Qu.:74.50   3rd Qu.:68.75   3rd Qu.:75.04
##  Max.   :90.88   Max.   :91.04   Max.   :80.75   Max.   :90.04
##      t_ws5           t_ws6           t_ws7           t_ws8
##  Min.   :27.00   Min.   :22.12   Min.   :23.38   Min.   :25.46
##  1st Qu.:51.58   1st Qu.:48.09   1st Qu.:50.21   1st Qu.:51.19
##  Median :64.81   Median :62.46   Median :63.79   Median :64.58
##  Mean   :62.83   Mean   :60.79   Mean   :61.83   Mean   :62.77
##  3rd Qu.:75.46   3rd Qu.:74.55   3rd Qu.:75.12   3rd Qu.:75.83
##  Max.   :89.04   Max.   :88.29   Max.   :86.88   Max.   :92.88
##      t_ws9            t_ws10          t_ws11          t_ws12
##  Min.   : 6.792   Min.   :22.62   Min.   :18.83   Min.   :22.83
##  1st Qu.:41.750   1st Qu.:49.84   1st Qu.:46.91   1st Qu.:49.29
##  Median :55.500   Median :63.94   Median :60.88   Median :62.38
##  Mean   :53.659   Mean   :61.94   Mean   :59.56   Mean   :61.07
##  3rd Qu.:67.250   3rd Qu.:75.54   3rd Qu.:73.88   3rd Qu.:74.83
##  Max.   :81.917   Max.   :88.71   Max.   :90.62   Max.   :90.38
##      t_ws13          t_ws14          t_ws15          t_ws16
##  Min.   :19.46   Min.   :18.79   Min.   :16.96   Min.   :25.83
##  1st Qu.:48.28   1st Qu.:46.58   1st Qu.:46.41   1st Qu.:51.92
##  Median :62.21   Median :60.44   Median :60.29   Median :63.62
##  Mean   :60.57   Mean   :58.93   Mean   :58.96   Mean   :62.70
##  3rd Qu.:74.50   3rd Qu.:72.75   3rd Qu.:72.47   3rd Qu.:75.75
##  Max.   :88.58   Max.   :88.54   Max.   :88.83   Max.   :86.08
```

```
##      t_ws17          t_ws18          t_ws19          t_ws20
##  Min.   :19.50   Min.   :24.04   Min.   :20.17   Min.   :24.46
##  1st Qu.:47.26   1st Qu.:52.12   1st Qu.:46.92   1st Qu.:52.11
##  Median :61.38   Median :65.44   Median :60.96   Median :64.35
##  Mean   :59.91   Mean   :63.28   Mean   :59.50   Mean   :63.15
##  3rd Qu.:74.21   3rd Qu.:76.00   3rd Qu.:73.67   3rd Qu.:76.21
##  Max.   :89.67   Max.   :88.21   Max.   :88.54   Max.   :85.96
##      t_ws21          t_ws22          t_ws23          t_ws24
##  Min.   :18.08   Min.   :17.50   Min.   :23.29   Min.   :21.58
##  1st Qu.:45.49   1st Qu.:45.50   1st Qu.:50.96   1st Qu.:47.50
##  Median :59.27   Median :59.35   Median :64.52   Median :61.62
##  Mean   :57.87   Mean   :58.17   Mean   :62.28   Mean   :60.49
##  3rd Qu.:71.67   3rd Qu.:72.26   3rd Qu.:75.42   3rd Qu.:74.62
##  Max.   :85.00   Max.   :88.21   Max.   :86.75   Max.   :91.29
##      t_ws25          t_ws26          t_ws27          t_ws28
##  Min.   :20.58   Min.   :22.75   Min.   :20.62   Min.   :20.96
##  1st Qu.:49.29   1st Qu.:48.95   1st Qu.:47.99   1st Qu.:47.21
##  Median :63.50   Median :63.44   Median :62.50   Median :61.69
##  Mean   :61.66   Mean   :61.73   Mean   :60.90   Mean   :59.98
##  3rd Qu.:75.14   3rd Qu.:75.79   3rd Qu.:75.33   3rd Qu.:74.08
##  Max.   :90.33   Max.   :90.79   Max.   :90.21   Max.   :88.50
##      rh_ws1           rh_ws2           rh_ws3          rh_ws4
##  Min.   : 22.42   Min.   : 7.167   Min.   :25.25   Min.   :19.92
##  1st Qu.: 60.71   1st Qu.:60.896   1st Qu.:60.65   1st Qu.:54.16
##  Median : 69.67   Median :69.354   Median :71.21   Median :65.08
##  Mean   : 69.22   Mean   :67.843   Mean   :69.89   Mean   :64.51
##  3rd Qu.: 78.26   3rd Qu.:77.042   3rd Qu.:79.58   3rd Qu.:74.71
##  Max.   :100.00   Max.   :96.000   Max.   :97.62   Max.   :97.50
##      rh_ws5           rh_ws6          rh_ws7          rh_ws8
##  Min.   : 23.96   Min.   :29.75   Min.   :30.50   Min.   : 26.08
##  1st Qu.: 63.28   1st Qu.:63.38   1st Qu.:65.79   1st Qu.: 60.77
##  Median : 71.83   Median :71.83   Median :74.21   Median : 69.79
##  Mean   : 70.87   Mean   :70.61   Mean   :72.87   Mean   : 69.12
##  3rd Qu.: 80.25   3rd Qu.:78.75   3rd Qu.:81.46   3rd Qu.: 78.88
##  Max.   :100.00   Max.   :97.00   Max.   :98.58   Max.   :100.00
##      rh_ws9          rh_ws10          rh_ws11         rh_ws12
##  Min.   : 13.83   Min.   : 24.00   Min.   :17.71   Min.   :18.58
##  1st Qu.: 55.65   1st Qu.: 61.58   1st Qu.:52.16   1st Qu.:52.45
##  Median : 68.50   Median : 71.60   Median :62.50   Median :63.12
##  Mean   : 66.82   Mean   : 70.78   Mean   :63.04   Mean   :63.26
##  3rd Qu.: 78.54   3rd Qu.: 80.88   3rd Qu.:73.43   3rd Qu.:74.26
##  Max.   :100.00   Max.   :100.00   Max.   :99.83   Max.   :98.25
##     rh_ws13         rh_ws14         rh_ws15          rh_ws16
##  Min.   :16.21   Min.   :20.38   Min.   : 15.96   Min.   :39.00
##  1st Qu.:57.79   1st Qu.:56.07   1st Qu.: 58.08   1st Qu.:67.74
##  Median :67.33   Median :65.67   Median : 68.42   Median :76.92
##  Mean   :66.25   Mean   :64.97   Mean   : 68.12   Mean   :75.27
##  3rd Qu.:75.74   3rd Qu.:74.10   3rd Qu.: 78.46   3rd Qu.:83.98
##  Max.   :99.71   Max.   :98.50   Max.   :100.00   Max.   :99.58
##     rh_ws17         rh_ws18         rh_ws19          rh_ws20
##  Min.   :20.58   Min.   :27.21   Min.   : 10.58   Min.   :17.79
##  1st Qu.:56.21   1st Qu.:64.42   1st Qu.: 59.02   1st Qu.:67.37
##  Median :66.02   Median :72.25   Median : 69.92   Median :76.02
##  Mean   :65.35   Mean   :71.01   Mean   : 68.67   Mean   :73.96
```

```
##  3rd Qu.:74.83   3rd Qu.:78.88   3rd Qu.: 79.38   3rd Qu.:82.04
##  Max.   :98.33   Max.   :96.75   Max.   :100.00   Max.   :98.04
##     rh_ws21          rh_ws22         rh_ws23         rh_ws24
##  Min.   : 17.83   Min.   :13.83   Min.   :16.21   Min.   : 20.54
##  1st Qu.: 57.91   1st Qu.:52.83   1st Qu.:66.71   1st Qu.: 59.75
##  Median : 68.40   Median :65.00   Median :75.21   Median : 69.38
##  Mean   : 66.97   Mean   :63.74   Mean   :73.88   Mean   : 68.91
##  3rd Qu.: 76.67   3rd Qu.:75.71   3rd Qu.:82.62   3rd Qu.: 78.58
##  Max.   :100.00   Max.   :95.12   Max.   :99.58   Max.   :100.00
##     rh_ws25          rh_ws26          rh_ws27         rh_ws28
##  Min.   : 23.12   Min.   : 30.46   Min.   :17.33   Min.   :22.04
##  1st Qu.: 58.92   1st Qu.: 62.45   1st Qu.:55.88   1st Qu.:61.61
##  Median : 67.71   Median : 71.58   Median :65.25   Median :70.58
##  Mean   : 67.69   Mean   : 71.28   Mean   :64.95   Mean   :69.86
##  3rd Qu.: 76.92   3rd Qu.: 80.58   3rd Qu.:74.17   3rd Qu.:78.42
##  Max.   :100.00   Max.   :100.00   Max.   :96.71   Max.   :98.58
```

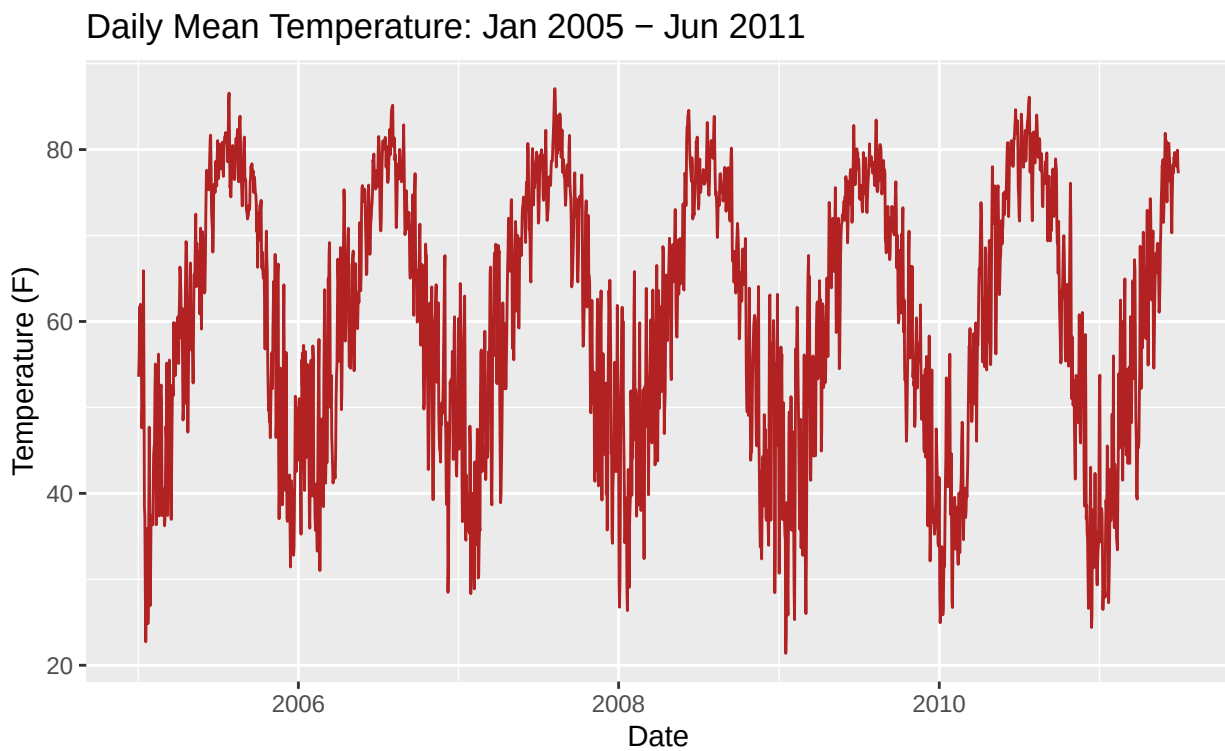
4. Exploratory Plots

```r
ggplot(daily_data, aes(x = Date, y = daily_mean_load)) +
  geom_line(color = "steelblue") +
  labs(title = "Daily Electricity Demand: Jan 2005 - Jun 2011",
       y = "Daily Mean Load (Wh)", x = "Date")
```

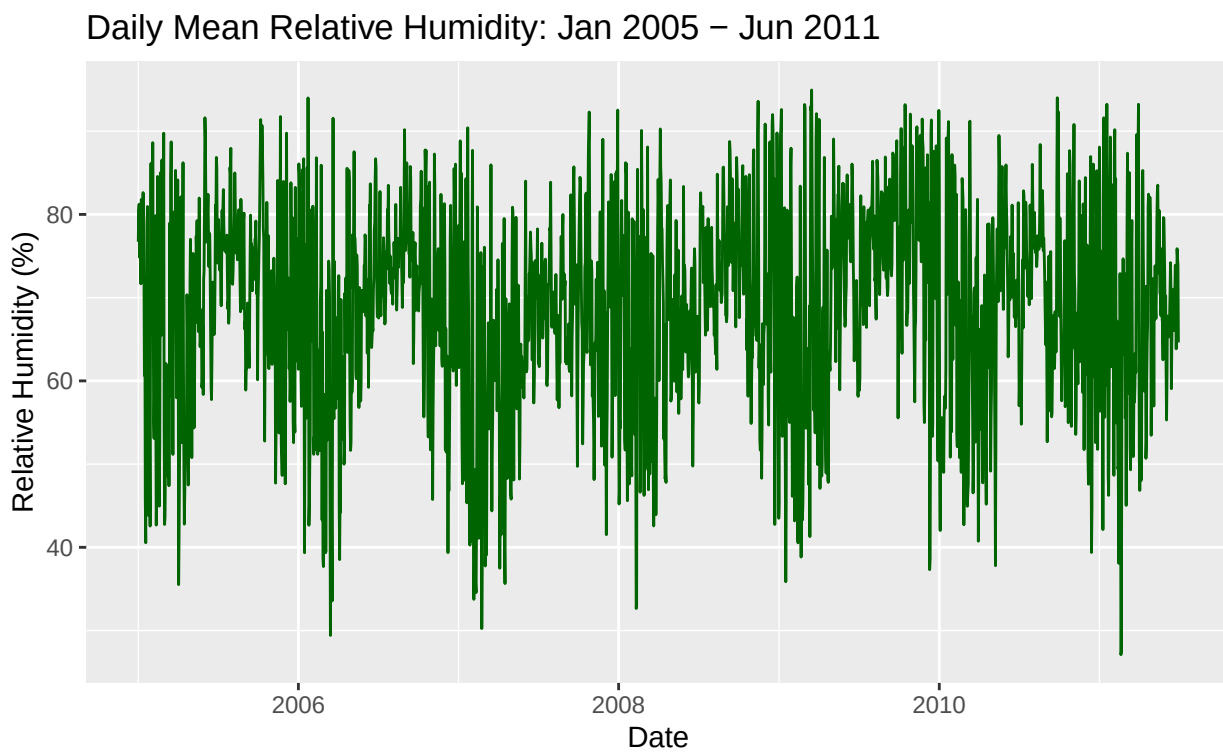



```r
ggplot(daily_data, aes(x = Date, y = daily_mean_temp)) +
  geom_line(color = "firebrick") +
```

```r
  labs(title = "Daily Mean Temperature: Jan 2005 - Jun 2011",
       y = "Temperature (F)", x = "Date")
```



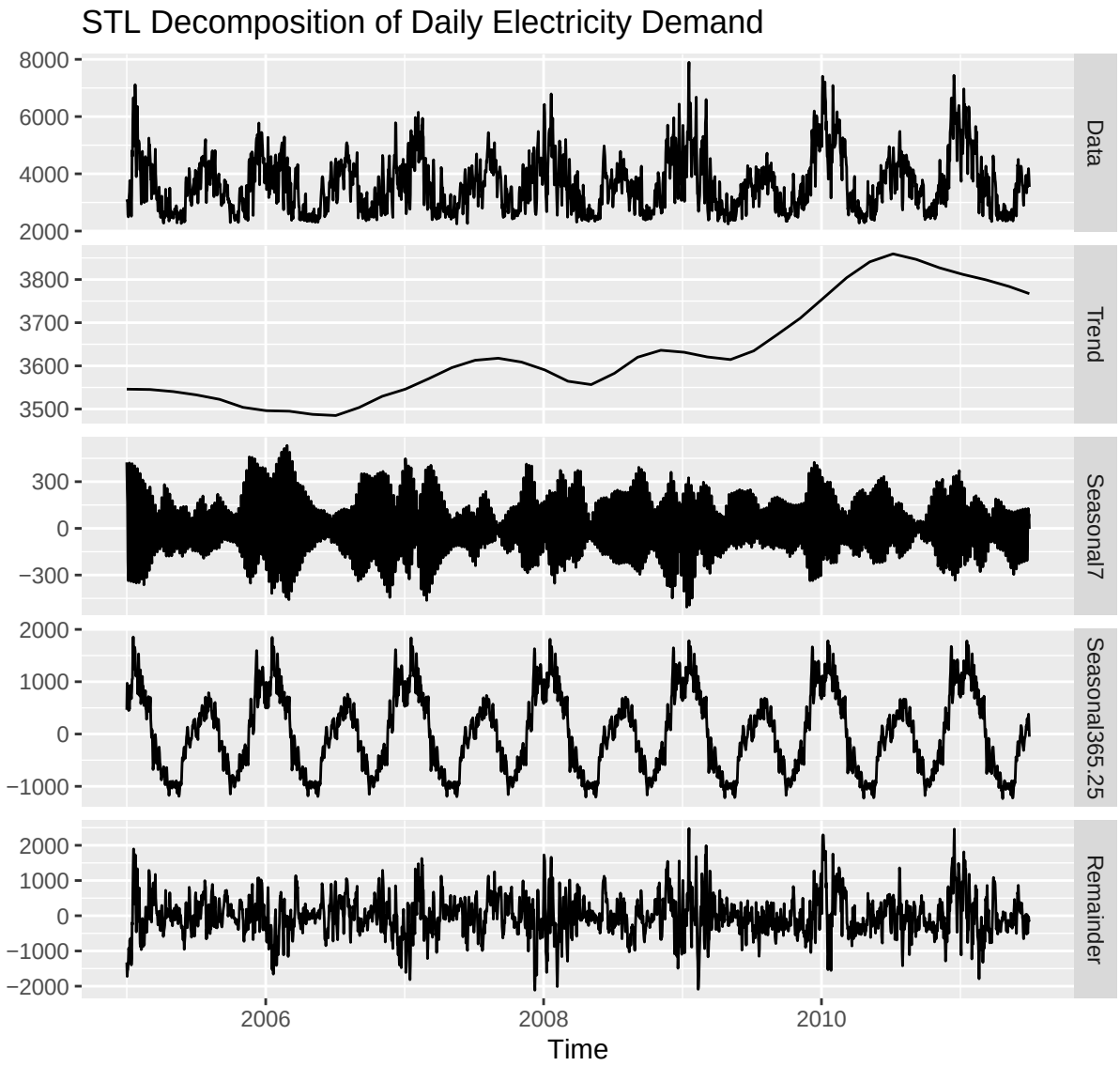

```r
ggplot(daily_data, aes(x = Date, y = daily_mean_rh)) +
  geom_line(color = "darkgreen") +
  labs(title = "Daily Mean Relative Humidity: Jan 2005 - Jun 2011",
       y = "Relative Humidity (%)", x = "Date")
```



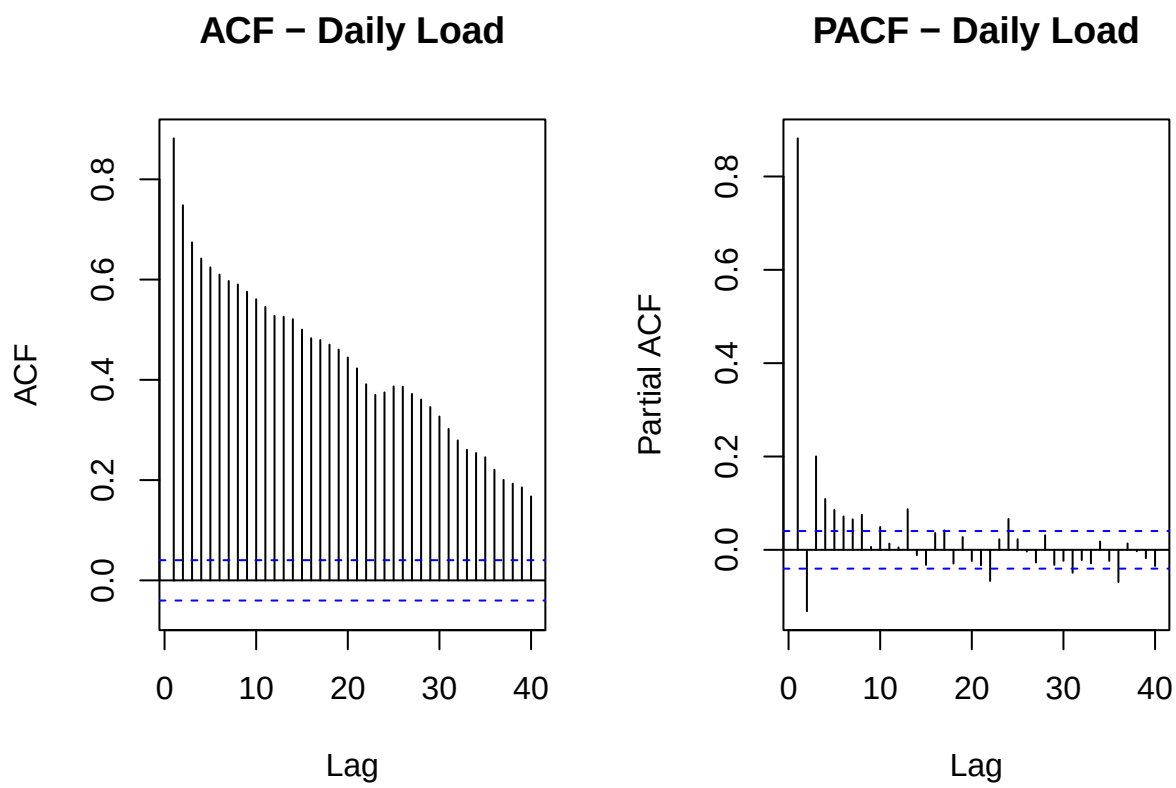
5. Time Series Decomposition and Autocorrelation

```
ts_daily_load <- msts(daily_data$daily_mean_load,
                      seasonal.periods = c(7, 365.25),
                      start = c(2005, 1))

ts_daily_load %>%
  mstl() %>%
  autoplot() +
  ggtitle("STL Decomposition of Daily Electricity Demand")
```




```r
par(mfrow = c(1, 2))
Acf(daily_data$daily_mean_load,  lag.max = 40, main = "ACF - Daily Load")
Pacf(daily_data$daily_mean_load, lag.max = 40, main = "PACF - Daily Load")
```




```
par(mfrow = c(1, 1))
```

6. Train / Validation / Full-Data Splits

Training: Jan 1, 2005 - Jun 30, 2010
Validation: Jul 1 - Jul 31, 2010 (31 days, mirrors the forecast horizon)
Full data: Jan 1, 2005 - Jun 30, 2011 (used for final refitting before submission)

```
train_data <- df_final %>% filter(Date <= as.Date("2010-06-30"))
val_data   <- df_final %>% filter(Date >= as.Date("2010-07-01") &
                                  Date <= as.Date("2010-07-31"))
full_data  <- df_final %>% filter(Date <= as.Date("2011-06-30"))

july_dates_2011 <- seq(as.Date("2011-07-01"), as.Date("2011-07-31"), by = "day")

cat("Split sizes - Train:", nrow(train_data),
    "| Val:", nrow(val_data),
    "| Full:", nrow(full_data), "\n")
```

```
## Split sizes - Train: 2007 | Val: 31 | Full: 2372
```

7. Future Regressor Construction (M1 only)

Temperature and humidity for July 2011 are estimated using a recency-weighted average of historical July values, giving more weight to recent years.

```
historical_julys <- df_final %>%
  filter(month(Date) == 7, year(Date) < 2011) %>%
  mutate(
    day_of_month = day(Date),
    year_weight  = year(Date) - 2004   # 2005 -> 1, ..., 2010 -> 6
  )

july_2011_features <- historical_julys %>%
  group_by(day_of_month) %>%
  summarise(across(
    .cols = where(is.numeric) & !contains("load") & !contains("year_weight"),
    .fns  = ~ weighted.mean(.x, w = year_weight, na.rm = TRUE)
  )) %>%
  ungroup() %>%
  mutate(Date = as.Date(paste0("2011-07-", str_pad(day_of_month, 2, pad = "0")))) %>%
  select(Date, everything(), -day_of_month)

cat("Future regressor rows:", nrow(july_2011_features), "\n")
```

```
## Future regressor rows: 31
```

8. Cross-Validation Setup

Walk-forward CV with 1-month assessment windows, 1-month skip, and cumulative (expanding) training window. Two CV objects are created: a 5-year initial window (M1, M2, M3, M5) and a 4-year initial window (M4).

```
# 5-year initial window: M1, M2, M3, M5
resamples_5yr <- time_series_cv(
  data       = train_data,
  date_var   = Date,
  initial    = "5 years",
  assess     = "1 month",
  skip       = "1 month",
  cumulative = TRUE
)

# 4-year initial window: M4
resamples_4yr <- time_series_cv(
  data       = train_data,
  date_var   = Date,
  initial    = "4 years",
  assess     = "1 month",
  #skip       = "1 month",
  cumulative = TRUE
)

cat("CV folds (5-year):", nrow(resamples_5yr), "\n")
```

```
## CV folds (5-year): 5
```

```r
cat("CV folds (4-year):", nrow(resamples_4yr), "\n")
```

```
## CV folds (4-year): 516
```

9. Recipes

```r
# M1: with temperature and humidity regressors
rec_with_reg <- recipe(
  daily_mean_load ~ Date + daily_mean_temp + daily_mean_rh,
  data = train_data
)

# M2, M3, M4, M5: date only
rec_no_reg <- recipe(
  daily_mean_load ~ Date,
  data = train_data
)
```

10. Tuning Grid

Tuned models (M1, M2, M3) share the same hyperparameter search space. The full grid below was used in the original extensive sweep. It is commented out here to allow the document to knit in reasonable time; the two-row active grid contains the best-performing combinations identified from that sweep.

```r
# ----------------------------------------------------------
# FULL GRID (used in original extensive hyperparameter sweep)
# Uncomment to reproduce the full search (~300 combinations)
# ----------------------------------------------------------
# prophet_grid_full <- expand.grid(
#   prior_scale_changepoints = c(0.001, 0.005, 0.01, 0.05, 0.1, 0.5),
#   prior_scale_seasonality  = c(0.1, 0.5, 1, 3, 5, 8, 10, 15, 20),
#   prior_scale_holidays     = c(0.1, 1.0, 10.0)
# )
# cat("Full grid size:", nrow(prophet_grid_full), "\n")

# ----------------------------------------------------------
# ACTIVE GRID (reduced for knitting; two combinations
# from the full sweep above, selected by CV MAPE)
# ----------------------------------------------------------
prophet_grid <- tibble(
  prior_scale_changepoints = c(0.005, 0.05),
  prior_scale_seasonality  = c(3.0,   10.0),
  prior_scale_holidays     = c(1.0,   10.0)
)

cat("Active grid size:", nrow(prophet_grid), "\n")
```

```
## Active grid size: 2
```

```r
print(prophet_grid)
```

```
## # A tibble: 2 x 3
##   prior_scale_changepoints prior_scale_seasonality prior_scale_holidays
##                      <dbl>                   <dbl>                <dbl>
## 1                    0.005                       3                    1
## 2                    0.05                       10                   10
```

---

11. Model Fitting

M1: Prophet with Regressors — Tuned (5-year CV, MAPE)

Additive seasonal model with temperature and humidity regressors. Hyperparameters were originally identified via RMSE and are re-evaluated here on MAPE for consistency with the competition metric.

```r
spec_m1_tune <- prophet_reg(
  seasonality_yearly       = TRUE,
  seasonality_weekly       = TRUE,
  prior_scale_changepoints = tune(),
  prior_scale_seasonality  = tune(),
  prior_scale_holidays     = tune()
) %>%
  set_engine("prophet",
             n.changepoints    = 22,
             changepoint.range = 0.72,
             seasonality.mode  = "additive")

wf_m1 <- workflow() %>%
  add_model(spec_m1_tune) %>%
  add_recipe(rec_with_reg)

cat("Tuning M1 (", nrow(prophet_grid), "combos x",
    nrow(resamples_5yr), "folds)...\n")
```

```
## Tuning M1 ( 2 combos x 5 folds)...
```

```r
tune_m1 <- tune_grid(
  wf_m1,
  resamples = resamples_5yr,
  grid      = prophet_grid,
  metrics   = metric_set(mape),
  control   = control_grid(verbose = FALSE, save_pred = TRUE)
)

best_m1 <- select_best(tune_m1, metric = "mape")
cat("M1 best params:\n"); print(best_m1)
```

```
## M1 best params:
```

```
## # A tibble: 1 x 4
##   prior_scale_changepoints prior_scale_seasonality prior_scale_holidays .config
##                      <dbl>                   <dbl>                <dbl> <chr>
## 1                    0.005                       3                    1 pre0_mo~
```

```
fit_m1 <- finalize_workflow(wf_m1, best_m1) %>% fit(train_data)
cat("M1 fitted.\n")
```

```
## M1 fitted.
```

M2: Prophet without Regressors — Tuned (5-year CV, RMSE)

No-regressor tuned model. Same spec and grid as M1 but without weather regressors. Best parameters selected by **RMSE** (original metric used before switching to MAPE).

```
spec_m2_tune <- prophet_reg(
  seasonality_yearly       = TRUE,
  seasonality_weekly       = TRUE,
  prior_scale_changepoints = tune(),
  prior_scale_seasonality  = tune(),
  prior_scale_holidays     = tune()
) %>%
  set_engine("prophet",
             n.changepoints    = 28,
             changepoint.range = 0.72,
             seasonality.mode  = "additive")

wf_m2 <- workflow() %>%
  add_model(spec_m2_tune) %>%
  add_recipe(rec_no_reg)

cat("Tuning M2 (", nrow(prophet_grid), "combos x",
    nrow(resamples_5yr), "folds)...\n")
```

```
## Tuning M2 ( 2 combos x 5 folds)...
```

```
tune_m2 <- tune_grid(
  wf_m2,
  resamples = resamples_5yr,
  grid      = prophet_grid,
  metrics   = metric_set(rmse, mape),
  control   = control_grid(verbose = FALSE, save_pred = TRUE)
)

best_m2 <- select_best(tune_m2, metric = "rmse")   # selected by RMSE
cat("M2 best params:\n"); print(best_m2)
```

```
## M2 best params:

## # A tibble: 1 x 4
##   prior_scale_changepoints prior_scale_seasonality prior_scale_holidays .config
##                      <dbl>                   <dbl>                <dbl> <chr>
## 1                    0.005                       3                    1 pre0_mo~
```

```r
fit_m2 <- finalize_workflow(wf_m2, best_m2) %>% fit(train_data)
cat("M2 fitted.\n")
```

```
## M2 fitted.
```

M3: Prophet without Regressors — Tuned v2 (5-year CV, MAPE)

Identical to M2 in every way except best parameters are selected by **MAPE** instead of RMSE. This is the updated submission reflecting the competition's target metric.

```r
spec_m3_tune <- prophet_reg(
  seasonality_yearly       = TRUE,
  seasonality_weekly       = TRUE,
  prior_scale_changepoints = tune(),
  prior_scale_seasonality  = tune(),
  prior_scale_holidays     = tune()
) %>%
  set_engine("prophet",
             n.changepoints    = 18,
             changepoint.range = 0.65,
             seasonality.mode  = "additive")

wf_m3 <- workflow() %>%
  add_model(spec_m3_tune) %>%
  add_recipe(rec_no_reg)

cat("Tuning M3 (", nrow(prophet_grid), "combos x",
    nrow(resamples_5yr), "folds)...\n")
```

```
## Tuning M3 ( 2 combos x 5 folds)...
```

```r
tune_m3 <- tune_grid(
  wf_m3,
  resamples = resamples_5yr,
  grid      = prophet_grid,
  metrics   = metric_set(rmse, mape),
  control   = control_grid(verbose = FALSE, save_pred = TRUE)
)

best_m3 <- select_best(tune_m3, metric = "mape")   # selected by MAPE
cat("M3 best params:\n"); print(best_m3)
```

```
## M3 best params:

## # A tibble: 1 x 4
##   prior_scale_changepoints prior_scale_seasonality prior_scale_holidays .config
##                      <dbl>                   <dbl>                <dbl> <chr>
## 1                    0.005                       3                    1 pre0_mo~
```

```r
fit_m3 <- finalize_workflow(wf_m3, best_m3) %>% fit(train_data)
cat("M3 fitted.\n")
```

```
## M3 fitted.
```

M4: Prophet without Regressors — Tuned (4-year CV, MAPE)

Additive seasonal model, no regressors, tuned via grid search. Uses a shorter 4-year CV window. Best parameters selected by MAPE.

```r
spec_m4_tune <- prophet_reg(
  seasonality_yearly       = TRUE,
  seasonality_weekly       = TRUE,
  prior_scale_changepoints = tune(),
  prior_scale_seasonality  = tune(),
  prior_scale_holidays     = tune()
) %>%
  set_engine("prophet",
             n.changepoints    = 28,
             changepoint.range = 0.72,
             seasonality.mode  = "additive")

wf_m4 <- workflow() %>%
  add_model(spec_m4_tune) %>%
  add_recipe(rec_no_reg)

cat("Tuning M4 (", nrow(prophet_grid), "combos x",
    nrow(resamples_4yr), "folds)...\n")
```

```
## Tuning M4 ( 2 combos x 516 folds)...
```

```r
tune_m4 <- tune_grid(
  wf_m4,
  resamples = resamples_4yr,
  grid      = prophet_grid,
  metrics   = metric_set(rmse, mape),
  control   = control_grid(verbose = FALSE, save_pred = TRUE)
)

best_m4 <- select_best(tune_m4, metric = "mape")
cat("M4 best params:\n"); print(best_m4)
```

```
## M4 best params:
```

```
## # A tibble: 1 x 4
##   prior_scale_changepoints prior_scale_seasonality prior_scale_holidays .config
##                      <dbl>                   <dbl>                <dbl> <chr>
## 1                    0.005                       3                    1 pre0_mo~
```

```r
fit_m4 <- finalize_workflow(wf_m4, best_m4) %>% fit(train_data)
cat("M4 fitted.\n")
```

```
## M4 fitted.
```

M5: Prophet without Regressors — Tuned (5-year CV, MAPE)

Additive seasonal model, no regressors, tuned via grid search. Uses the longer 5-year CV window. Best parameters selected by MAPE.

```r
spec_m5_tune <- prophet_reg(
  seasonality_yearly       = TRUE,
  seasonality_weekly       = TRUE,
  prior_scale_changepoints = tune(),
  prior_scale_seasonality  = tune(),
  prior_scale_holidays     = tune()
) %>%
  set_engine("prophet",
             n.changepoints    = 28,
             changepoint.range = 0.72,
             seasonality.mode  = "additive")

wf_m5 <- workflow() %>%
  add_model(spec_m5_tune) %>%
  add_recipe(rec_no_reg)

cat("Tuning M5 (", nrow(prophet_grid), "combos x",
    nrow(resamples_5yr), "folds)...\n")
```

```
## Tuning M5 ( 2 combos x 5 folds)...
```

```r
tune_m5 <- tune_grid(
  wf_m5,
  resamples = resamples_5yr,
  grid      = prophet_grid,
  metrics   = metric_set(rmse, mape),
  control   = control_grid(verbose = FALSE, save_pred = TRUE)
)

best_m5 <- select_best(tune_m5, metric = "mape")
cat("M5 best params:\n"); print(best_m5)
```

```
## M5 best params:
```

```
## # A tibble: 1 x 4
##   prior_scale_changepoints prior_scale_seasonality prior_scale_holidays .config
##                      <dbl>                   <dbl>                <dbl> <chr>
## 1                    0.005                       3                    1 pre0_mo~
```

```r
fit_m5 <- finalize_workflow(wf_m5, best_m5) %>% fit(train_data)
cat("M5 fitted.\n")
```

```
## M5 fitted.
```

---

12. Calibration and Validation Accuracy (July 2010 Holdout)

```r
model_table <- modeltime_table(
  fit_m1,   # M1: Prophet + regressors, tuned
  fit_m2,   # M2: Prophet no regressors, tuned
  fit_m3,   # M3: Prophet no regressors, tuned v2
  fit_m4,   # M4: Prophet no regressors, baseline 4yr CV
  fit_m5    # M5: Prophet no regressors, baseline 5yr CV
)

calibration_table <- model_table %>%
  modeltime_calibrate(new_data = val_data)

cat("\n-- Validation Accuracy (July 2010 holdout) --\n")
```

```
##
## -- Validation Accuracy (July 2010 holdout) --
```

```r
modeltime_accuracy(calibration_table) %>%
  select(.model_id, .model_desc, rmse, mae, mape, rsq) %>%
  mutate(Submission = c(
    "Submission_Prophet_Tuned",
    "Sub_M4_Prophet_NoReg_Baseline",
    "Sub_M4_Prophet_NoReg_Baseline(v2)",
    "Submission_Prophet_Tuned_NEW",
    "Submission_Prophet_Tuned_Newest51"
  )) %>%
  kable(digits = 2, caption = "Validation Accuracy: July 2010 Holdout") %>%
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 2: Validation Accuracy: July 2010 Holdout

.model_id	.model_desc	rmse	mae	mape	rsq	Submission
1	PROPHET W/ REGRESSORS	780.10	659.19	16.02	0.21	Submission_Prophet_Tuned
2	PROPHET	613.83	512.28	12.58	0.07	Sub_M4_Prophet_NoReg_Baseline
3	PROPHET	613.83	512.28	12.58	0.07	Sub_M4_Prophet_NoReg_Baseline(v2)
4	PROPHET	613.83	512.28	12.58	0.07	Submission_Prophet_Tuned_NEW
5	PROPHET	613.83	512.28	12.58	0.07	Submission_Prophet_Tuned_Newest51

---

13. Final Refit on Full Data and Forecasting July 2011

```r
cat("Refitting all models on full dataset (Jan 2005 - Jun 2011)...\n")
```

```
## Refitting all models on full dataset (Jan 2005 - Jun 2011)...
```

```r
refit_table <- calibration_table %>%
  modeltime_refit(data = full_data)

# M1 requires future regressors; M2-M5 only need dates
future_with_reg <- july_2011_features
future_no_reg   <- tibble(Date = july_dates_2011)

# Forecast M1 separately (requires regressors)
fcst_m1 <- refit_table %>%
  filter(.model_id == 1) %>%
  modeltime_forecast(
    new_data    = future_with_reg,
    actual_data = full_data
  )

# Forecast M2-M5 together (no regressors)
fcst_m2_m5 <- refit_table %>%
  filter(.model_id %in% c(2, 3, 4, 5)) %>%
  modeltime_forecast(
    new_data    = future_no_reg,
    actual_data = full_data
  )

# Extract prediction vectors
preds_m1 <- fcst_m1    %>% filter(.key == "prediction") %>% pull(.value)
preds_m2 <- fcst_m2_m5 %>% filter(.model_id == 2, .key == "prediction") %>% pull(.value)
preds_m3 <- fcst_m2_m5 %>% filter(.model_id == 3, .key == "prediction") %>% pull(.value)
preds_m4 <- fcst_m2_m5 %>% filter(.model_id == 4, .key == "prediction") %>% pull(.value)
preds_m5 <- fcst_m2_m5 %>% filter(.model_id == 5, .key == "prediction") %>% pull(.value)

cat("Forecast lengths - M1:", length(preds_m1), "M2:", length(preds_m2),
    "M3:", length(preds_m3), "M4:", length(preds_m4), "M5:", length(preds_m5), "\n")
```

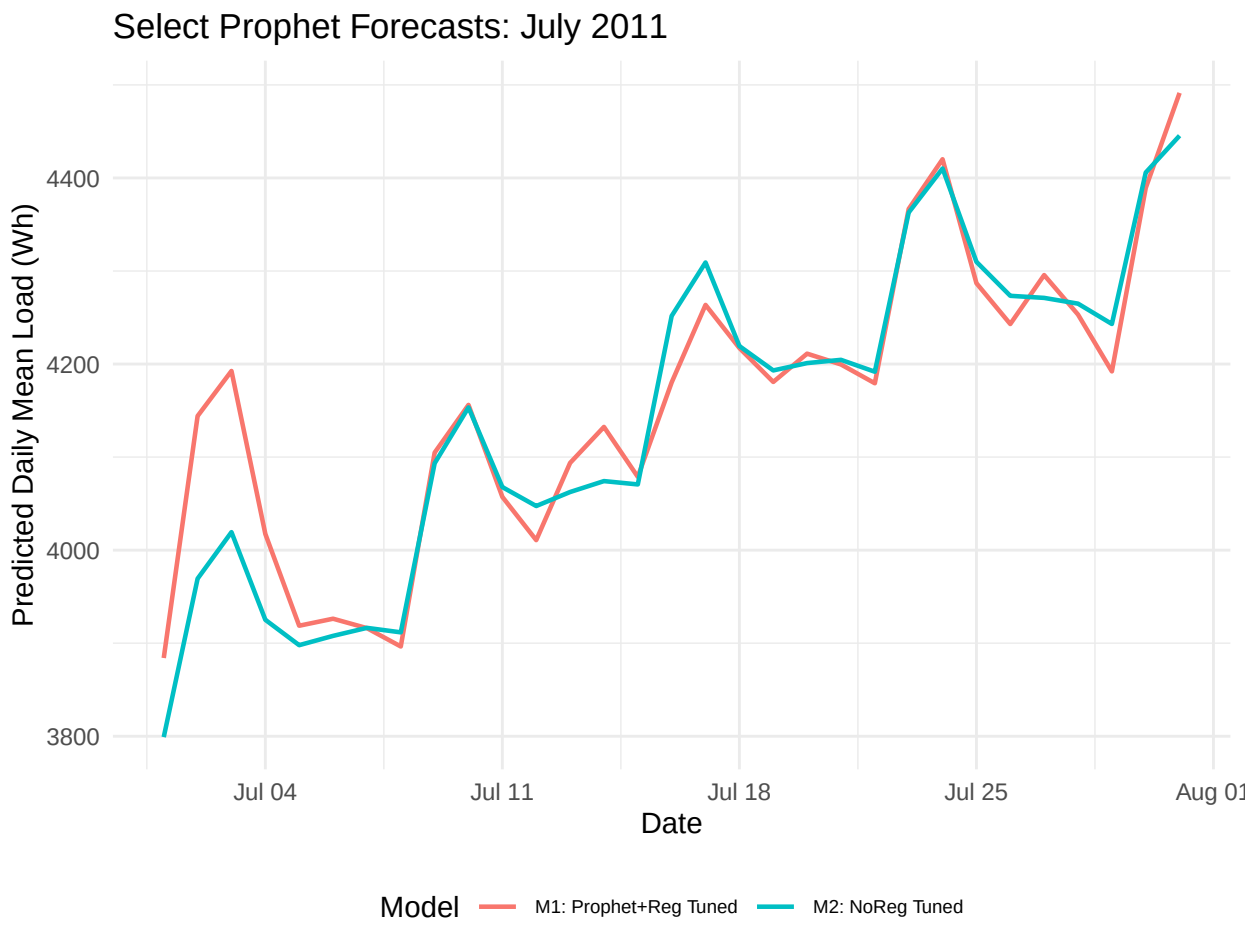
```
## Forecast lengths - M1: 31 M2: 31 M3: 31 M4: 31 M5: 31
```

---

14. Forecast Visualization

```r
tibble(
  Date                            = july_dates_2011,
  `M1: Prophet+Reg Tuned`         = preds_m1,
  `M2: NoReg Tuned`               = preds_m2,
```

```r
) %>%
  pivot_longer(-Date, names_to = "Model", values_to = "Prediction") %>%
  ggplot(aes(x = Date, y = Prediction, color = Model)) +
  geom_line(linewidth = 0.8) +
  labs(title = "Select Prophet Forecasts: July 2011",
       x = "Date", y = "Predicted Daily Mean Load (Wh)", color = "Model") +
  theme_minimal() +
  theme(legend.position = "bottom",
        legend.text = element_text(size = 7))
```



```r
save_sub <- function(preds_vec, filename) {
  tibble(
    date = format(july_dates_2011, "%Y-%m-%d"),
    load = round(preds_vec, 2)
  ) %>%
    write_csv(file.path("./Submissions", filename))
  cat("Saved:", filename, "\n")
}

dir.create("./Submissions", showWarnings = FALSE)
```

```r
save_sub(preds_m1, "Submission_Prophet_Tuned.csv")
```

```
## Saved: Submission_Prophet_Tuned.csv
```

```r
save_sub(preds_m2, "Sub_M4_Prophet_NoReg_Baseline.csv")
```

```
## Saved: Sub_M4_Prophet_NoReg_Baseline.csv
```

```r
save_sub(preds_m3, "Sub_M4_Prophet_NoReg_Baseline_v2.csv")
```

```
## Saved: Sub_M4_Prophet_NoReg_Baseline_v2.csv
```

```r
save_sub(preds_m4, "Submission_Prophet_Tuned_NEW.csv")
```

```
## Saved: Submission_Prophet_Tuned_NEW.csv
```

```r
save_sub(preds_m5, "Submission_Prophet_Tuned_Newest51.csv")
```

```
## Saved: Submission_Prophet_Tuned_Newest51.csv
```

```r
cat("\nAll 5 submissions saved to ./Submissions/\n")
```

```
##
## All 5 submissions saved to ./Submissions/
```